

Technical Document #779: Database Systems - Comprehensive Overview

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Relational databases organize data into tables with rows and columns. SQL is used to query and manipulate structured data.

Database migration tools manage schema changes and data transformations. Flyway and Liquibase are popular database migration tools.

Database monitoring tracks performance metrics and identifies bottlenecks. Query analysis and indexing strategies optimize database performance.

NoSQL databases handle unstructured and semi-structured data. Document stores, key-value stores, and graph databases are common NoSQL types.

ACID properties (Atomicity, Consistency, Isolation, Durability) ensure reliable transaction processing in databases.

Database connection pooling reuses database connections to reduce overhead. It improves performance for applications with many concurrent users.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Key Takeaways:

- Relational databases organize data into tables with rows and columns.
- Database monitoring tracks performance metrics and identifies bottlenecks.
- Database connection pooling reuses database connections to reduce overhead.